Many find this conclusion troubling or even downright unacceptable. Einstein was one. In one of physics' most time-honored utterances, Einstein admonished the quantum stalwarts that "God does not play dice with the Universe." He felt that probability was turning up in fundamental physics because of a subtle version of the reason it turns up at the roulette wheel: some basic incompleteness in our understanding. The universe, in Einstein's view, had no room for a future whose precise form involves an element of chance. Physics should predict how the universe evolves, not merely the likelihood that any particular evolution might occur. But experiment after experiment —some of the most convincing ones being carried out after his death —convincingly confirm that Einstein was wrong. As the British theoretical physicist Stephen Hawking has said, on this point "Einstein was confused, not the quantum theory."6

Nevertheless, the debate about what quantum mechanics really means continues unabated. Everyone agrees on how to use the equations of quantum theory to make accurate predictions. But there is no consensus on what it really means to have probability waves, nor on how a particle "chooses" which of its many possible futures to follow, nor even on whether it really does choose or instead splits off like a branching tributary to live out all possible futures in an ever-expanding arena of parallel universes. These interpretational issues are worthy of a book-length discussion in their own right, and, in fact, there are many excellent books that espouse one or another way of thinking about quantum theory. But what appears certain is that no matter how you interpret quantum mechanics, it undeniably shows that the universe is founded on principles that, from the standpoint of our day-to-day experiences, are bizarre.

The meta-lesson of both relativity and quantum mechanics is that when we deeply probe the fundamental workings of the universe we may come upon aspects that are vastly different from our expectations. The boldness of asking deep questions may require unforeseen flexibility if we are to accept the answers.

Feynman's Perspective

Richard Feynman was one of the greatest theoretical physicists since Einstein. He fully accepted the probabilistic core of quantum mechanics, but in the years following World War II he offered a